



4AG.023 High Temperature Felts

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Final Advanced Materials offers felts and blankets resistant to high temperatures up to 2,500 °C. These products are mainly used for thermal and sound insulation for the manufacture of industrial furnaces, or as a component of thermal protection. Many industries use felts for their manufacturing processes. With varying densities and thicknesses, some references can be aluminized to improve their thermal properties.

Applications

- High temperature protection
- Electrical, thermal, acoustic insulation
- Alternative to asbestos and ceramic fibres
- Catalytic heaters
- Industrial furnace insulation



High Temperature Felts

Comparative Tables

Properties	Units	Basalt	E-Glass	Silicate	Pure Silica	Biosoluble	Ceramic	Graphite	Zirconia
Density	kg/m ³	120 to 137.5	100 to 160	120 to 170	20 & 130	64 to 128	288	80	0.24 to 0.64
Continuous Temp.	°C	600	550	1,000	1,000	1,000	1,260	500	2,000
Peak Temp.	°C	700	840	1,200	1,200	1,200	1,760	2,000	2,500

Product range

Basalt Needled Felts, up to 700 °C

Made from basalt fibres of 8 to 16 µm, the felts are classified M0 according to European standard EN 13-501-1. Their temperature resistance covers from – 280 °C to 700 °C. They do not burn, do not melt, produce neither smoke nor toxic gases, and are environmentally friendly and recyclable. They are mainly used as electrical, thermal, and acoustic insulators. In addition, basalt fibre is the most environmentally friendly high temperature resistant material when it comes to both manufacturing and recycling it.

Applications:

- High temperature protection
- Electrical insulation
- Thermal insulation

Technical data:

Property	Unit	Basalt Needled Felts		
Item N°		014-0100	014-0110	014-0120
Area density	g/m ²	480	780	1,100
Density	kg/m ³	120	130	137.5
Thickness	mm	4	6	8
Length	m	400	400	400
Operating Temperature	°C	600	600	600
Peak Temperature	°C	700	700	700

E-glass Fibre Felts, up to 700 °C

E-glass fibres are the most widespread in use, whether in the textile industry or for composite materials. This high-quality fibre is distinguished by its resistance to high temperatures up to 550 °C and its excellent electrical insulating properties. This fibre is rot-proof, resistant to the most common chemical agents and dimensionally stable, even in the event of substantial variations in humidity and temperature.



High Temperature Felts

This felt has excellent mechanical, chemical and dielectric properties. In addition, it can be reinforced with a layer of aluminium to withstand radiant heat up to 1,650 °C.

Applications:

- Acoustic insulation
- Thermal insulation

Technical data:

Property	Unit	E-Glass Fibre Felts
Item N°		207-4701
Density	kg/m ³	100 to 160
Thickness	mm	3 to 25
Area Density	g/m ²	300 – 4,000
Width	mm	1,000 to 1,500
Diameter of the Fibre	µm	approx. 6-9
Thermal Conductivity	W.m ⁻¹ .K ⁻¹	0.03
Electrical Rigidity	kV/mm	60 to 100
Operating Temperature	°C	550
Peak Temperature	°C	700

Other dimensions available on request.

Silicate Fibre Felts, up to 1,200 °C

These felts are made of more than 98 % silica (S98) and have a silicate fibre base that allows them to withstand application up to 1 200 °C. Asbestos-free and inorganic, these products do not contain toxic substances and are not irritating. Thanks to their good thermal and mechanical properties, they are excellent alternatives to ceramic fibre-based products.

Silicate fibre felts do not contain any organic binders. The fibres are mechanically bonded, which makes the felt compact, flexible and gives it a good hold during processing. S98 fibre felts are used in the shipbuilding and automotive industries as heat and acoustic insulation and as a component for thermal and fire protection.

Applications:

- Heat insulation
- Acoustic insulation
- Component for thermal and fire protection

Technical data:

Property	Unit	Silicate Fibre Felts
Item N°		207-8701
Density	kg/m ³	120 - 170
Width	mm	1,000
Thickness	mm	3 to 25



High Temperature Felts

Property		Unit	Silicate Fibre Felts
Roll Length		mm	10 to 75
Continuous Temperature		°C	1,000
Peak temperature		°C	1,200
Thermal Conductivity 120 kg/m ³	200 °C	W.m ⁻¹ .K ⁻¹	0.053
	400 °C		0.089
	600 °C		0.119
	800 °C		0.175
	1,000 °C		0.190

Other dimensions available on request.

Ultra-Pure Silica Felts, up to 1,200 °C

Our ultra-pure silica felts contain 99.95 % SiO₂. Composed of silica fibres assembled with polyvinyl alcohol, they can be used in temperatures of up to 1,200 °C. Even at a continuous temperature of 1,000 °C, they do not shrink and retain a very low thermal conductivity and exceptional mechanical properties.

Our felts are particularly popular in the aerospace industry for insulating engines thanks to their outstanding insulating and fireproof characteristics as well as their resistance to vibrations and their very long life. Mainly used between two refractory sheets, we also offer flexible solutions made with coated fabrics. Their outstanding properties are also useful for domestic applications such as catalytic heaters and industrial furnace insulation.

Applications:

- Insulation of industrial furnaces
- Catalytic supports for heating systems
- Replacement of asbestos and ceramic fibre
- Thermal insulation in the manufacture of optical fibre
- Engine insulation in aeronautics

Technical Data:

Property	Unit	Low Density Felt	High Density Felt
Item N°		190-0100	190-0110
Area Density	g/m ²	80	200
Density	kg/m ³	~ 20	130
Thickness	mm	~ 8	2
Width	mm	1,000	1,000
Length	mm	3,500	25,000
Operating Temperature	°C	1,000	1,000
Peak Temperature	°C	1,200	1,200
Thermal Conductivity At 500 °C	W.m ⁻¹ .K ⁻¹	0.274	0.107

Custom-cut available for low density felts.



High Temperature Felts

Biosoluble Fibres Felts, up to 1,200 °C

Biosoluble fibre felts are obtained from long fibres of alkaline earth silicate. With high thermal stability, they retain a flexible fibrous structure up to a temperature of 1,200 °C. Their high-temperature insulation performance is remarkable.

This range of product contains no binders or lubricants and emits no smoke or odour when heated. Biosoluble fibre felts are flexible and easy to cut or install. Biosoluble fibre blankets are available laminated with an aluminium film to improve the thermal properties of the sheets.

Biosoluble fibres felts' composition excludes all carcinogenic classification as per Q in directive 97/69 EC, unlike pure silica and silicate fibres. Biosoluble fibres felts are an ideal solution to replace ceramic fibres.

Applications:

- Chimney, furnace and pipe insulation
- Heating and furnace lining
- Heat shield
- Component in the manufacture of thermal protection

Technical data:

Property		Unit	Biosoluble Fibres Felts			
Density		kg/m ³	64	80	96	128
Tensile Strength		kPa	75	75	75	75
Linear Shrinkage after 4 hrs at 1,200 °C		%	1	1	1	1
Operating Temperature		°C	1,000	1,000	1,000	1,000
Peak Temperature		°C	1,200	1,200	1,200	1,200
Thermal Conductivity	200 °C	W.m ⁻¹ .K ⁻¹	0.06	0.06	0.05	0.05
	400 °C		0.11	0.09	0.09	0.08
	600 °C		0.18	0.15	0.14	0.12
	800 °C		0.29	0.24	0.21	0.18
	1,000 °C		0.42	0.36	0.29	0.25

Soft Carbon & Graphite Felts, up to 2,000 °C

Carbon and graphite felts with thicknesses of 6 and 12 mm with a width of 1,200 and 1,270 mm are available. Specific widths can be provided and special shapes can be cut to your specifications.

Soft carbon and graphite felts offer many advantages. They have low thermal conductivity which leads to energy savings of up to 75 %, high resistivity (coupling in an inductive field only occurs above 12 kHz) and do not charge with electrostatic energy. They are thermally stable in oxidizing atmospheres up to 350 °C, in protective or vacuum atmospheres up to approximately 2,000 °C for graphite and 1,000 °C for carbon. Their composition is very pure with very little sulphur and ash content.



High Temperature Felts

Applications:

- Thermal insulation for vacuum furnaces and inert gas furnaces
- Heated smelting and holding furnaces
- Inductively heated furnaces
- Filters for hot and/or corrosive gases and liquids, for molten metals
- Backing strips for soldering and welding

Technical data:

Property		Unit	Carbon Felt		Graphite Felt	
Item N°			KFA 5	KFA 10	GFA 5	GFA 10
Area Density		g/m ²	540	1,100	500	1,000
Thickness		mm	6	12	6	12
Width		mm	1,200	1,270	1,200	1,270
Length		m	30	30	15-30	15-30
Ash Content		%	1.7	1.7	0.04	0.04
Operating Temp.		°C	1,000	1,000	2,000	2,000
Thermal Conductivity	at 200 °C	W.m ⁻¹ .K ⁻¹	0.05	0.05	0.03	0.03
	at 600 °C		0.09	0.09	0.17	0.17
	at 1,000 °C		1.6	1.6	0.6	0.6

GFA5 and GFA10 references are to cut according to your needs.

Rigid Graphite Felt, up to 2,000 °C

Rigid graphite felt is available with a thickness of 40 mm (max. size 1,524 x 1,219 mm). This product is made of graphite fibres and a carbon binder. It is suitable for temperatures up to 2,200 °C. Specific widths and shapes can be cut.

Applications:

- Thermal insulation
- Support element
- Component of high-temperature furnaces

Technical data:

Property		Unit	Rigid graphite felt
Item N°			MFA
Density		g/cm ³	0.2
Thickness		mm	40
Width		mm	1,219
Length		mm	1,524
Compressive Strength		MPa	0.7
Flexural Strength		MPa	0.8
Ash Content		ppm	1,000
Operating Temperature		°C	2,000
Thermal Conductivity	at 200 °C	W.m ⁻¹ .K ⁻¹	0.2



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	at 600 °C		0.35
	at 1,000 °C		0.95

Rigid graphite felt is also available with an enduction on one or both faces
(carbon or graphite reinforcement)

Zirconium Oxide Fibre Felts, up to 2,500 °C

ZYF zirconium oxide fibre felts are the result of the Zircar Process by Zircar Zirconia which transforms an organic fibre into a ceramic fibre. The new fibre usually has a diameter of 6 to 10 µm and a serrated outer face. It can withstand up to 2,500 °C. Zirconium oxide fibres are made of nearly 100 % zirconia phase stabilized with 10 % yttria.

ZYF is a flexible, light weight, porous, needled felt. ZYF is an excellent all-purpose high temperature insulation that can be used in methods of crystal growing as iridium and platinum crucible insulation, and precious metal condensate reclamation material. These felts are inorganic and do not require organic binders.

ZYF-50, ZYF-100 et ZYF-150 :

- Exceptional thermal insulation even beyond 1,930 °C;
- Flexible up to 1,350 °C;
- Low outgassing and odor even at high temperature;
- Light and porous structure;
- Low thermal conductivity;
- High thermal stability ;
- High resistance to corrosive, oxidizing and reducing atmospheres.

Other versions have been treated to enhance their properties and are also available:

ZYF-100A :

- Similar to the ZYF-100 version but with a lower level of sodium.

ZYF-A2.13 :

- Similar to the ZYF-150 version but harder
- Filled with alumina powder and high fired to a rigid board.

ZYF-S.75G :

- Similar to the ZYF-100 version but more flexible
- Filled with 0.75 g of silica powder per 1 g of ZYF, remaining pliable.

ZYF-Z1.5G :

- Similar to the ZYF-100 version
- Filled with 1.5 g of zirconia powder per 1 g of ZYF and dyed green for identification

Applications

ZYF-50, ZYF-100 et ZYF-150 :

- All-purpose
- High temperature setter
- High temperature shield and cover
- High temperature insulation
- Crystal growth



High Temperature Felts

ZYF-100A :

- All-purpose
- Glass melting
- High temperature setter
- High temperature shield and cover
- High temperature insulation
- Crystal growth
- Electrolytic cell gasket

ZYF-A2.13 :

- Diffusion burner membrane for fuel fired heaters installed in long-haul trucks and high end passenger cars
- Setter plate for vacuum sintering flat sheets of powdered metal materials

ZYF-S.75G :

- Gasket material in fuel cells

ZYF-Z1.5G :

- Gasket material in fuel cells

Technical data:

Property	Unit	ZYF-50	ZYF-100	ZYF-100A	ZYF-150	ZYF-A2.13	ZYF-S.75G	ZYF-Z1.5G
Composition								
ZrO ₂ * + Y ₂ O ₃	Wt %	99 +	99+	99+	99+	97+	56+	99+
Al ₂ O ₃		< 0.01	< 0.01	-	< 0.01	2 ± 1	< 0.01	< 0.01
SiO ₂		< 0.02	< 0.02	-	< 0.02	< 0.02	43	< 0.01
Na ₂ O	ppm	-		< 500	-	-	-	-
Thickness	mm	1.27	2.54	2.54	3.81	4.06 ± 0.89	2.54 ± 0.76	2.54 ± 0.76
Bulk Porosity	%	96	96	96	96	94	88	89
Bulk Density	g/cm ³	0.24	0.24	0.24	0.24	0.32	0.48	0.64
Tensile Strength	g/cm	107	286	714	357	1,250	304	196
Area Density	g/m ²	342	635	-	1,123	1,025	1,074	1,514
Melting Point	°C	2,593	2,593	2,593	2,593			
Continuous Temperature	°C	2,000	2,000	2,000	2,000	1,600	1,600	1,600
Compressive Strength	Psi at 10 % compression	3	3	-	5	13	13	13
Min. wrapping diameter before breaking	mm	6.35	19.05	19.05	76.20	914.40	50.80	304.8
Heat capacity								
at 93 °C	J/kg/K	544	544	544	544	-	-	-
at 2,371 °C	J/kg/K	754	754	754	754	-	-	-
Vapour Pressure at 1,371 °C	Torr	8.10 ⁻¹²				-	-	-



High Temperature Felts

% Shrinkage after 1 hr at 1,679 °C	%	5	4	6	4	1.5	5	14
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*1 - 2 wt% hafnia occurs naturally with zirconia and does not affect performance.

Felt type ZYF can be reinforced with Zircar Zirconia cement type ZR-CEM or with Zircar Zirconia hardener type ZR-RIG. Other coatings or coatings are possible according to your needs.

Physical variables included in this documentation are provided by way of indication only and do not, under any circumstances, constitute a contractual undertaking. Please contact our technical service if you require any additional information.